Sound: pitch and amplitude

antocciphysics

UNIT 6 / DAY 8 / HONORS + SEI

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a longitudinal-wave model to distinguish pitch and sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pitch” and “amplitude” to compare your model or result. Sentence starters are optional.

pitch: the perceived highness or lowness of a sound

amplitude: maximum displacement from equilibrium for the wave

longitudinal: a disturbance parallel to the wave’s travel direction

01 Bring back an earlier idea

Light moves from a transparent medium with n=1.6 into air (n=1).
Find critical angle using $\sin\theta_c = 1/n$.
Describe incidence above it in the ideal model.

Word help: medium = the material through which a mechanical wave travels

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $T = 1/f$; $v = f\lambda$.
Higher f generally means higher pitch. Amplitude affects intensity under matched conditions, not frequency.

Worked reasoning: A tuning fork makes 440 cycles each second, so its frequency is 440 Hz. A louder version can have the same frequency but greater amplitude.

Your turn: Draw two displacement-time traces with equal periods and different amplitudes.

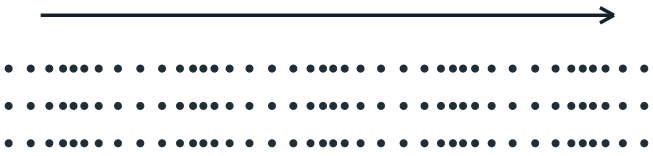
Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?

Sentence starter: I notice ... . This supports ... because ... .



Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Solve and justify

Two tones in same air have frequencies 250 and 500 Hz.
Use speed 340 m/s to find wavelengths. If pressure amplitude doubles at fixed medium, compare intensity using $I \propto A^2$.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Turning up a speaker’s volume does not normally double a song’s pitch.
Explain which wave property mainly changes.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Bodies initially at 340 K and 360 K exchange heat in an isolated system.
Explain what happens to total energy and to the temperature-difference opportunity for work at equilibrium.

Word help: heat = energy transferred because of a temperature difference

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A 500 Hz tone lasts .10 s.
Find number of cycles and period.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two tones have frequencies 400 and 405 Hz. Using beat frequency as their difference, find the beat rate.
Explain why beats are not a new 405-minus 400 Hz pitch from either source alone.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3